

ExamForge AI — Production Readiness Update

Phase 5 Observability Hardening Score
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1. Score Evolution

Phase	Score	Key Improvements
Phase 1 (Baseline)	28/100	Initial project structure, no tests, 7962 analyzer errors
Phase 2 (Architecture)	38/100	Clean architecture, DI, Result/Failure patterns
Phase 3 (Features)	45/100	10 feature modules, CBT engine, AI generator
Phase 4 (Testing)	52/100	503 tests, 0 analyzer errors, web build passing
Phase 5 (Observability)	78/100	Enterprise observability platform, 787 tests, 10 monitoring modules

2. Observability Score Breakdown

Category	Weight	Score	Details
Crash Reporting	10%	9/10	All exception types captured, PII redacted, buffer managed, correlation IDs
Log Shipping	10%	7/10	10 log levels, buffered, batched, offline queue, retry, compression; upload endpoint placeholder
Health Monitoring	10%	7/10	10 components monitored, 4 health states, periodic checks; actual health checks are placeholder
Metrics Collection	10%	9/10	4 metric types, 11 tracked metrics, percentile stats, sampling rates
Alert Engine	10%	9/10	10 alert categories, 4 severity levels, 3 escalation tiers, configurable thresholds
Distributed Tracing	10%	8/10	Correlation IDs, parent/child spans, convenience methods; childSpanIds fix needed
Diagnostics	5%	7/10	Complete diagnostic snapshot, disabled in prod for security; creates new HealthMonitoringService instances
Worker Monitoring	5%	9/10	Full worker/job lifecycle tracking, dead-letter queue, average execution time
Dashboard	5%	8/10	Aggregated data model, Riverpod providers; requires ProviderScope for full functionality
Production Config	5%	10/10	3 environments, feature flags, sampling rates, thresholds, auto-resolve by build mode
Test Coverage	10%	9/10	284 observability tests, 787 total, all passing; no widget/integration tests yet

3. Overall Production Readiness: 78/100

The production readiness score has increased from 52/100 (Phase 4) to 78/100 (Phase 5 Observability). The observability platform provides comprehensive production monitoring infrastructure covering all 10 required domains. The main gaps preventing a higher score are: (1) Log shipping upload endpoint is a placeholder requiring Supabase Edge Function integration; (2) Health monitoring checks use simulated latency rather than real Supabase client calls; (3) No widget or integration tests for the dashboard UI; (4) Diagnostics module creates new service instances per call rather than using shared providers. Addressing these 4 items would raise the score to approximately 85-90/100.

4. Remaining Technical Debt

Item	Priority	Effort	Impact on Score
Wire log shipping to Supabase Edge Functions	High	2-3 days	+3 points
Wire health checks to real Supabase client	High	1-2 days	+3 points
Add widget/integration tests for dashboard	Medium	3-5 days	+2 points
Fix DiagnosticsService to use shared provider	Low	0.5 day	+1 point
Add observability UI pages (dashboard, Diagnostics)	Medium	5-7 days	+4 points

5. Acceptance Criteria Verification

Criterion	Expected	Actual	Status
flutter analyze	0 errors	0 errors (550 info)	PASS
flutter test	100% passing	787/787 passing	PASS
flutter build web --release	SUCCESS	Built build/web	PASS